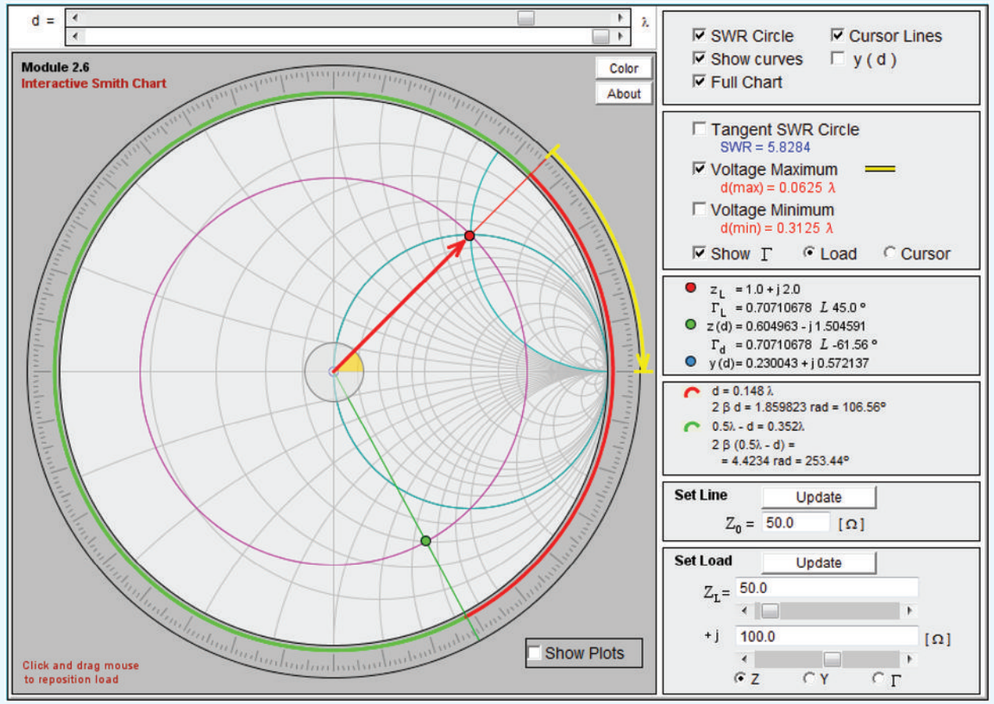
Module 2.6 Interactive Smith Chart Locate the load on the Smith chart; display the corresponding reflection coefficient and SWR circle; "move" to a new location at a distance d from the load; read the wave impedance $Z(d)$ and phase-shifted reflection coefficient Γ_d; perform impedance to admittance transformations and vice versa; and use all of these tools to solve transmission-line problems via the Smith chart.



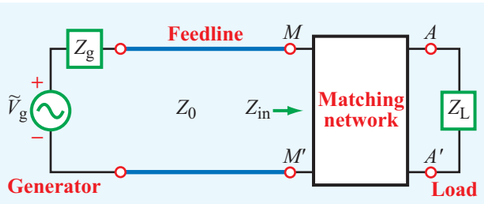
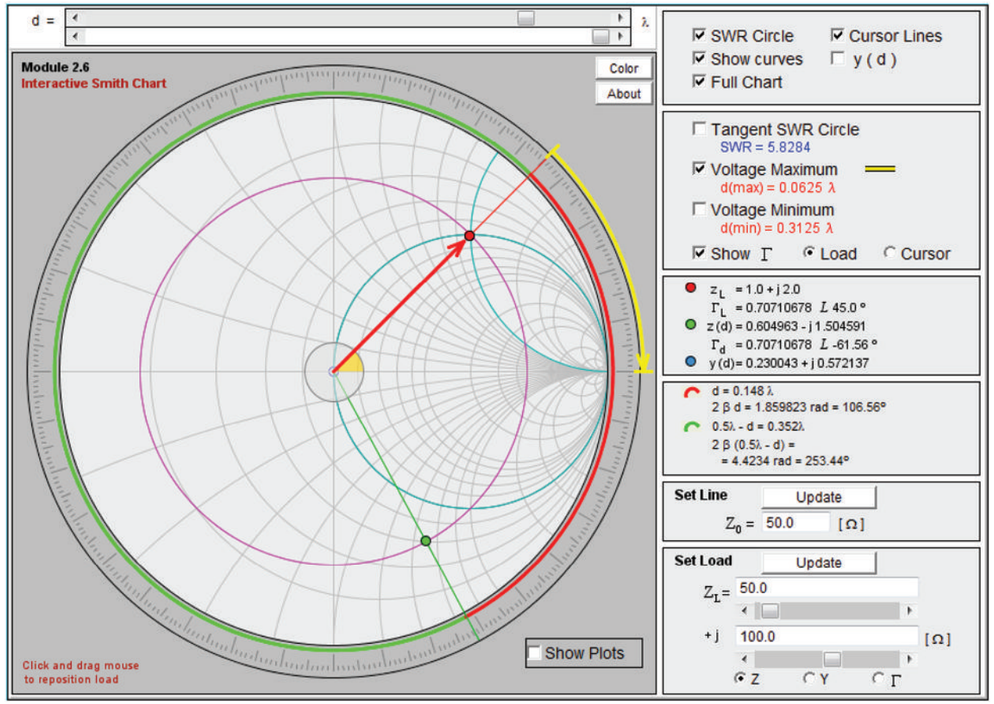


Figure 2-32 The function of a matching network is to transform the load impedance Z_L such that the input impedance Z_{in} looking into the network is equal to Z_0 of the feedline.

Within the matching network, reflections can occur at both terminals (AA' and MM'), creating a standing-wave pattern, but the net result (of all of the multiple reflections within the matching network) is that the wave incident from the source experiences no reflection when it reaches terminals MM'. This is achieved by designing the matching network to exhibit an impedance equal to Z_0 at MM' when looking into the network from the transmission line side. If the network is lossless, then all of the power going into it will end up in the load.

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模块 2.6 交互式史密斯圆图 在史密斯圆图上定位负载；显示相应的反射系数和驻波比圆； “移动”到距负载距离 d 的新位置；读取波阻抗Z(d)和相移反射系数Γ_d；执行阻抗到导纳的变换，反之亦然；并使用所有这些工具通过史密斯圆图解决传输线问题。



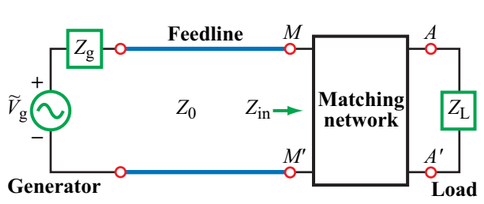
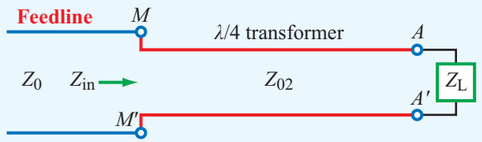
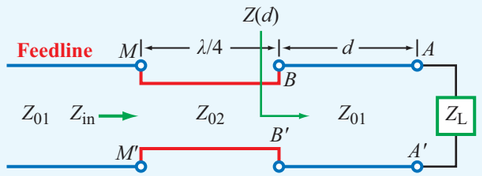


图2-32 匹配网络的功能是变换负载阻抗Z_L，使得网络中的输入阻抗Z_{in}等于馈线的Z_0。

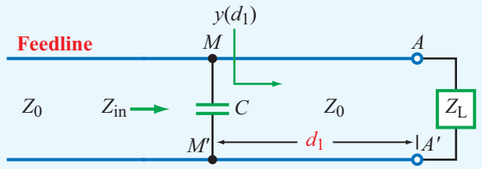
在匹配网络内，反射可能发生在两个终端（AA′和 MM′），从而产生驻波模式，但最终结果（匹配网络内的所有多次反射）是从源入射的波在到达终端 MM′时不会经历反射。这是通过设计匹配网络来实现的，当从传输线侧观察网络时，匹配网络在 MM′处表现出等于 Z_0 的阻抗。如果网络是无损的，那么进入网络的所有功率最终都会进入负载。



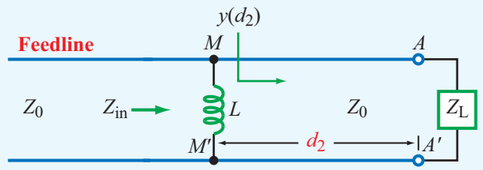
(a) If Z_L is real: in-series $\lambda/4$ transformer inserted at AA'



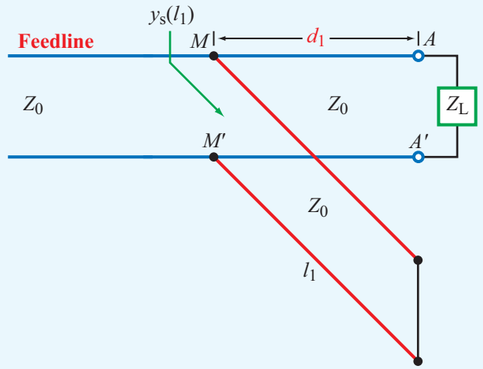
(b) If Z_L = complex: in-series $\lambda/4$ transformer inserted at $d = d_{max}$ or $d = d_{min}$



(c) In-parallel insertion of capacitor at distance d_1



(d) In-parallel insertion of inductor at distance d_2



(e) In-parallel insertion of a short-circuited stub

Figure 2-33 Five examples of in-series and in-parallel matching networks.

► Matching networks may consist of lumped elements, such as capacitors and inductors (but not resistors because resistors incur ohmic losses), or of sections of transmission lines with appropriate lengths and terminations. ◄

The matching network, which is intended to match a load impedance $Z_L = R_L + jX_L$ to a lossless transmission line with characteristic impedance Z_0, may be inserted either in series (between the load and the feedline) as in **Fig. 2-33(a)** and **(b)** or in parallel (**Fig. 2-33(c)** to **(e)**). In either case, the network has to transform the real part of the load impedance from R_L (at the load) to Z_0 at MM' in **Fig. 2-32** and transform the reactive part from X_L (at the load) to zero at MM'. To achieve these two transformations, the matching network must have at least two degrees of freedom (that is, two adjustable parameters).

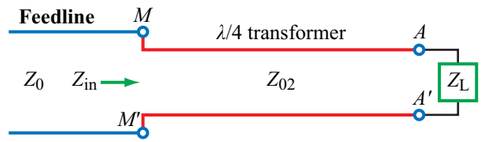
If $X_L = 0$, the problem reduces to a single transformation, in which case matching can be realized by inserting a quarter-wavelength transformer (Section 2-8.5) next to the load (**Fig. 2-33(a)**).

► For the general case where $X_L \neq 0$, a $\lambda/4$ transformer can be designed to provide the desired matching, but it has to be inserted at a distance d_{max} or d_{min} from the load (**Fig. 2-33(b)**), where d_{max} and d_{min} are the distances to voltage maxima and minima, respectively. ◄

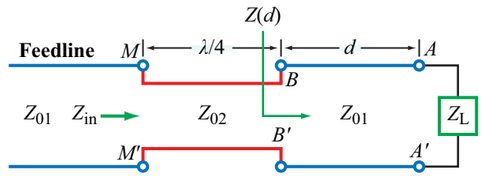
The design procedure is outlined in Module 2.7. The in-parallel insertion networks shown in **Fig. 2-33(c)–(e)** are the subject of Examples 2-13 and 2-14.

2-11.1 Lumped-Element Matching

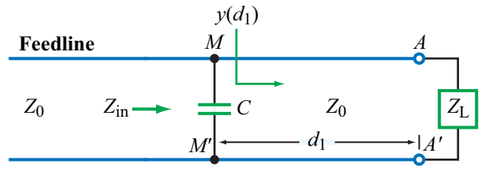
In the arrangement shown in **Fig. 2-34**, the matching network consists of a single lumped element, either a capacitor or an



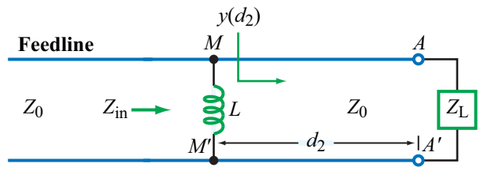
(a) If Z_L is real: in-series $\lambda/4$ transformer inserted at AA'



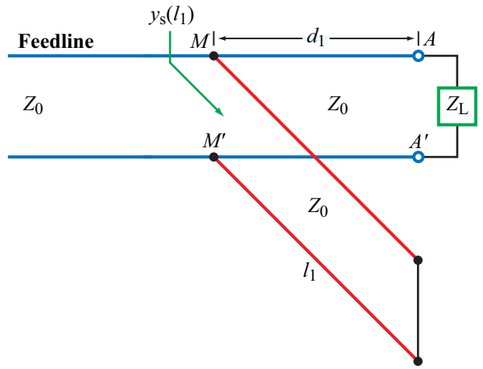
(b) If Z_L = complex: in-series $\lambda/4$ transformer inserted at $d = d_{max}$ or $d = d_{min}$



(c) In-parallel insertion of capacitor at distance d_1



(d) In-parallel insertion of inductor at distance d_2



(e) In-parallel insertion of a short-circuited stub

图2-33 串联和并联匹配网络的五个示例。

► 匹配网络可能由集总元件组成，例如电容器和电感器（但不是电阻器，因为电阻器会产生欧姆损耗），或者由具有适当长度和终端的传输线部分组成。 ◄

匹配网络的目的是将负载阻抗 $Z_L = R_L + jX_L$ 与具有特性阻抗 Z_0 的无损传输线相匹配，可以如图 2-33(a) 和 (b) 所示串联插入（在负载和馈线之间），也可以如图 2-33(c) 至 (e) 所示并联插入。无论哪种情况，网络都必须将负载阻抗的实部从负载)处的R_L (转换为图2-32中MM′处的Z_0，并将无功部分从负载)处的X_L (转换为MM′处的零。为了实现这两种变换，匹配网络必须至少具有两个自由度（即两个可调整的参数）。

如果 $X_L = 0$，则问题简化为单个变换，在这种情况下，可以通过插入来实现匹配

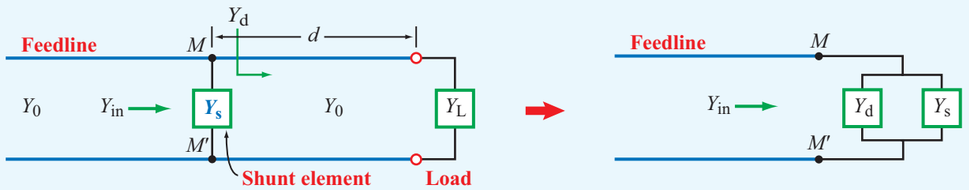
负载旁边的四分之一波长变压器（第 2-8.5 节）（图 2-33(a)）。

► 对于 $X_L \neq 0$ 的一般情况，可以设计一个 $\lambda/4$ 变压器来提供所需的匹配，但必须将其插入到距负载的距离 d_{max} 或 d_{min} 处（图 2-33(b)），其中 d_{max} 和 d_{min} 分别是到电压最大值和最小值的距离。 ◄

设计过程在模块 2.7 中概述。图 2-33(c)-(e) 中所示的并行插入网络是示例 2-13 和 2-14 的主题。

2-11.1 集总元件匹配

在图 2-34 所示的布置中，匹配网络由单个集总元件（电容器或电容器）组成。



(a) Transmission-line circuit (b) Equivalent circuit

Figure 2-34 Inserting a reactive element with admittance Y_s at MM' modifies Y_d to Y_{in}.

inductor, connected in parallel with the line at a distance d from the load. Parallel connections call for working in the admittance domain. Hence, the load is denoted by an admittance Y_L, and the line has a characteristic admittance Y_0. The shunt element has an admittance Y_s. At MM', Y_d is the admittance due to the transmission-line segment to the right of MM'. The input admittance Y_{in} (referenced at a point just to the left of MM') is equal to the sum of Y_d and Y_s:

$$Y_{in} = Y_d + Y_s. \tag{2.138}$$

In general, Y_d is complex, and Y_s is purely imaginary because it represents a reactive element (capacitor or inductor). Hence, Eq. (2.138) can be written as

$$Y_{in} = (G_d + jB_d) + jB_s = G_d + j(B_d + B_s). \tag{2.139}$$

When all quantities are normalized to Y_0, Eq. (2.139) becomes

$$y_{in} = g_d + j(b_d + b_s). \tag{2.140}$$

To achieve a matched condition at MM', it is necessary that $y_{in} = 1 + j0$, which translates into two specific conditions, namely

$$g_d = 1 \quad \textbf{(real-part condition)}, \tag{2.141a}$$
$$b_s = -b_d \quad \textbf{(imaginary-part condition)}. \tag{2.141b}$$

The real-part condition is realized through the choice of d, which is the distance from the load to the shunt element, and the imaginary-part condition is realized through the choice of lumped element (capacitor or inductor) and its value. These two choices are the two degrees of freedom needed in order to match the load to the feedline.

Example 2-13: Lumped Element

A load impedance $Z_L = 25 - j50\ \Omega$ is connected to a 50 Ω transmission line. Insert a shunt element to eliminate reflections towards the sending end of the line. Specify the insert location d (in wavelengths), the type of element, and its value given that $f = 100$ MHz.

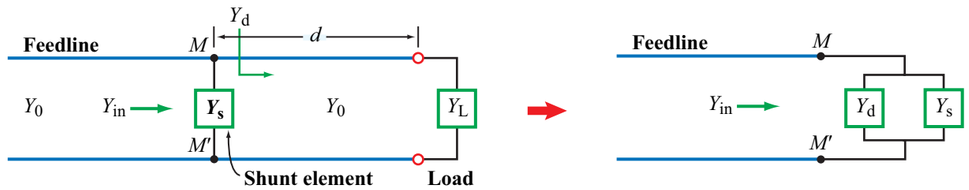
Solution: The normalized load impedance is

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1,$$

which is represented by point A on the Smith chart of **Fig. 2-35**. Next, we draw the constant S circle through point A. As alluded to earlier, to perform the matching task, it is easier to work with admittances than with impedances. The normalized load admittance y_L is represented by point B, which is obtained by rotating point A over 0.25λ or equivalently by drawing a line from point A through the chart center to the image of point A on the S circle. The value of y_L at B is

$$y_L = 0.4 + j0.8,$$

and it is located at position 0.115λ on the WTG scale. In the admittance domain, the r_L circles become g_L circles, and the x_L circles become b_L circles. To achieve matching, we need to move from the load toward the generator a distance d such that the normalized input admittance y_d of the line terminated in the load (**Fig. 2-34**) has a real part of 1. This condition is satisfied by either of the two ***matching points*** C and D on the Smith charts of **Figs. 2-35** and **2-36**, respectively, corresponding to intersections of the S circle with the $g_L = 1$ circle. Points C and D represent two possible solutions for the distance d in **Fig. 2-34(a)**.



(a) Transmission-line circuit (b) Equivalent circuit

图 2-34 在 MM′ 处插入导纳 Y_s 的电抗元件将 Y_d 修改为 Y_{in}。

电感器，与负载距离为 d 的线路并联。并行连接要求在准入域中工作。因此，负载用导纳 Y_L 表示，线路具有特征导纳 Y_0。分流元件的导纳为 Y_s。在 MM′ 处，Y_d 是 MM′ 右侧传输线段引起的导纳。以 MM′) 左侧点为参考的输入导纳 Y_{in} (等于 Y_d 和 Y_s 之和：

$$Y_{in} = Y_d + Y_s. \tag{2.138}$$

一般来说，Y_d 是复数，Y_s 是纯虚数，因为它代表电抗元件（电容器或电感器）。因此，等式。 (2.138) 可以写成

$$Y_{in} = (G_d + jB_d) + jB_s = G_d + j(B_d + B_s). \tag{2.139}$$

当所有量都归一化为 Y_0 时，方程： (2.139) 变为

$$y_{in} = g_d + j(b_d + b_s). \tag{2.140}$$

为了在 MM′ 处实现匹配条件，需要 $y_{in} = 1 + j0$，它转化为两个特定条件，即

$$g_d = 1 \quad \textbf{(real-part condition)}, \tag{2.141a}$$
$$b_s = -b_d \quad \textbf{(imaginary-part condition)}. \tag{2.141b}$$

实部条件通过选择 d（负载到分流元件的距离）来实现，虚部条件通过选择集总元件（电容器或电感器）及其值来实现。这两个选择是使负载与馈线相匹配所需的两个自由度。

例 2-13： 集总元件

负载阻抗Z_L =25−j50Ω连接到50Ω传输线。插入分流元件以消除线路发送端的反射。指定插入位置 d（以波长为单位）、元素类型及其值（假定 f = 100 MHz）。

解：归一化负载阻抗为

$$z_L = \frac{Z_L}{Z_0} = \frac{25 - j50}{50} = 0.5 - j1,$$

其由图2-35的史密斯圆图上的A点表示。接下来，我们通过 A 点绘制常数 S 圆。如前所述，为了执行匹配任务，使用导纳比使用阻抗更容易。归一化负载导纳 y_L 用 B 点表示，B 点是将 A 点旋转 0.25λ 或等效地从 A 点穿过图表中心到 S 圆上 A 点的图像画一条线得到的。 B 处 y_L 的值为

$$y_L = 0.4 + j0.8,$$

它位于WTG标尺上的位置0.115λ。在导纳域中，r_L圆变为g_L圆，x_L圆变为b_L圆。为了实现匹配，我们需要从负载向发电机移动距离 d，使得负载端接线路（图 2-34）的归一化输入导纳 y_d 的实部为 1。图 2-34 的史密斯圆图上的两个匹配点 C 和 D 中的任何一个都可以满足这个条件。2-35和2-36分别对应于S圆与g_L = 1圆的交点。 C点和D点代表图2-34(a)中距离d的两种可能的解。

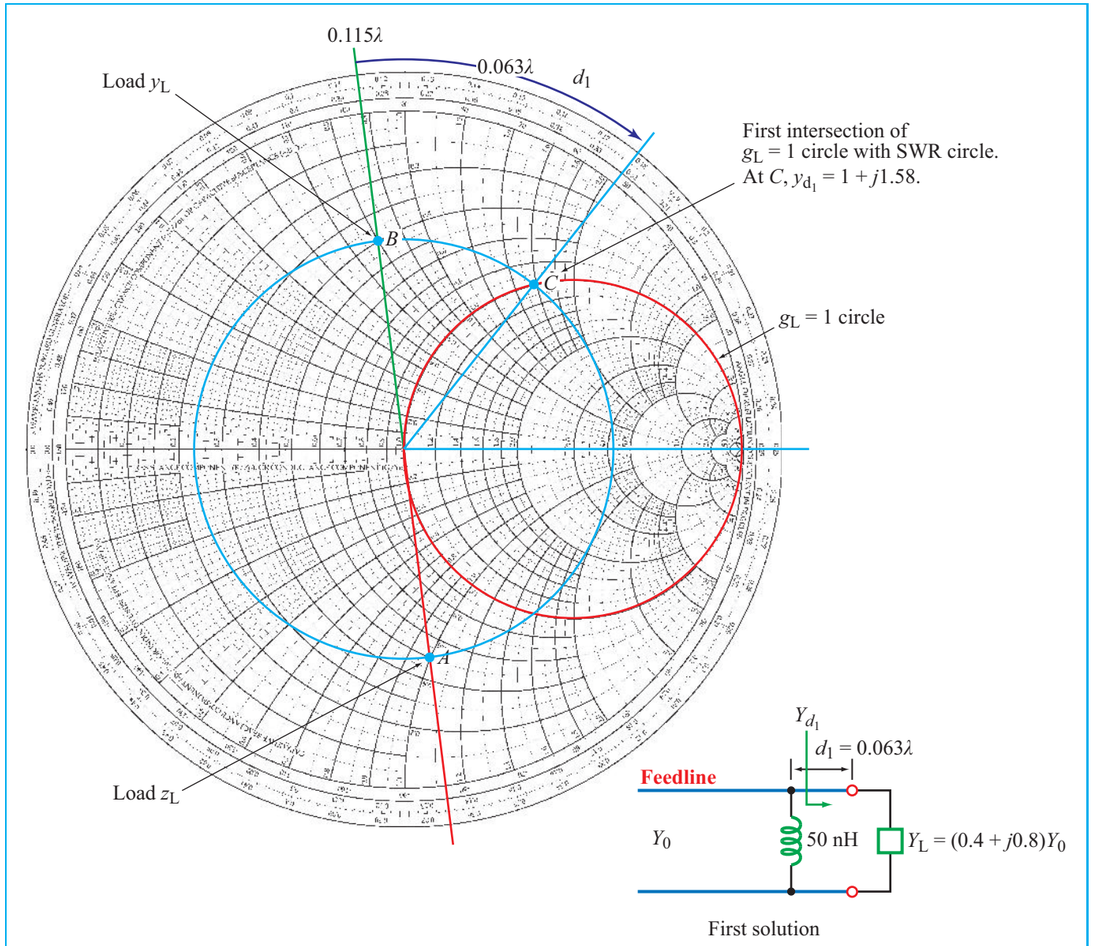


Figure 2-35 Solution for point C of Example 2-13. Point A is the normalized load with $z_L = 0.5 - j1$; point B is $y_L = 0.4 + j0.8$. Point C is the intersection of the SWR circle with the $g_L = 1$ circle. The distance from B to C is $d_1 = 0.063\lambda$.

Solution for Point C (Fig. 2-35): At C,

$$y_{d_1} = 1 + j1.58,$$

which is located at 0.178λ on the WTG scale. The distance between points B and C is

$$d_1 = (0.178 - 0.115)\lambda = 0.063\lambda.$$

Looking from the generator toward the parallel combination of the line connected to the load and the shunt element, the normalized input admittance at terminals MM' is

$$y_{in_1} = y_{s_1} + y_{d_1},$$

(continued on p. 101)

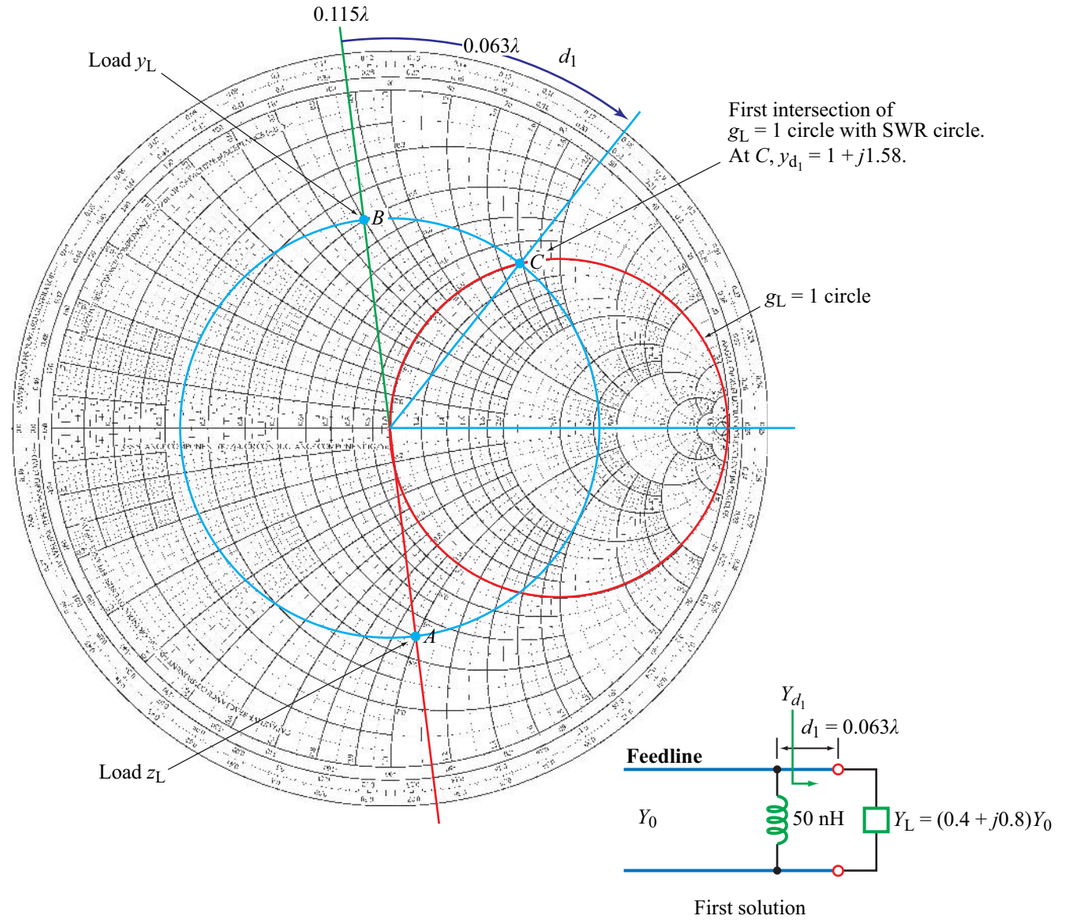


图 2-35 例 2-13 中 C 点的解法。 A点为$z_L = 0.5 - j1$的归一化载荷； B 点为 $y_L = 0.4 + j0.8$。 C 点是 SWR 圆与 $g_L = 1$ 圆的交点。 B 到 C 的距离为 $d_1 = 0.063\lambda$。

C 点的解（图 2-35）： 在 C 处，

$$y_{d_1} = 1 + j1.58,$$

它位于 WTG 尺度上的 0.178λ。 B 点和 C 点之间的距离为

$$d_1 = (0.178 - 0.115)\lambda = 0.063\lambda.$$

从发电机看向连接到负载的线路和并联元件的并联组合，端子 MM′ 处的归一化输入导纳为

$$y_{in_1} = y_{s_1} + y_{d_1},$$

(继续上页。 101)

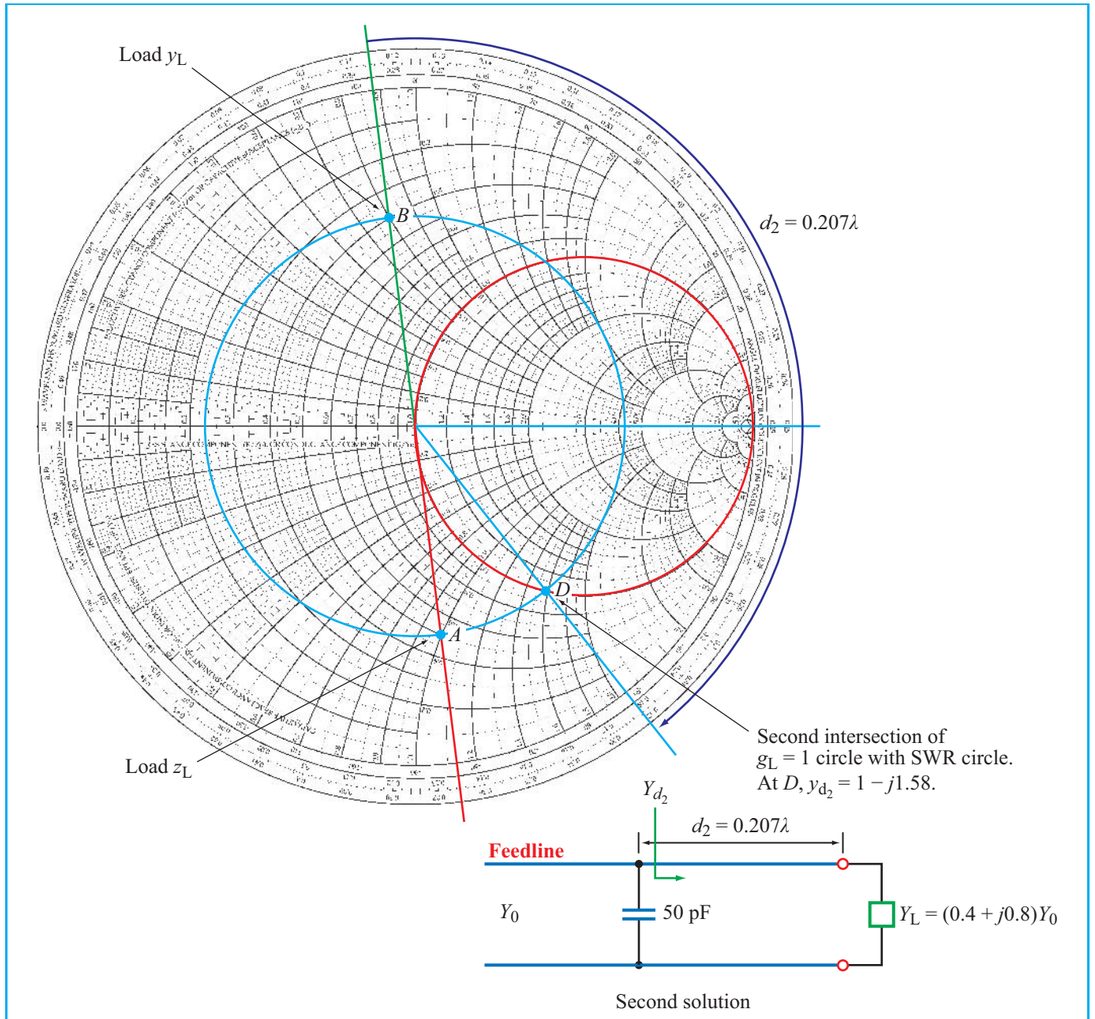


Figure 2-36 Solution for point D of Example 2-13. Point D is the second point of intersection of the SWR circle and the $g_L = 1$ circle. The distance B to D gives $d_2 = 0.207\lambda$.

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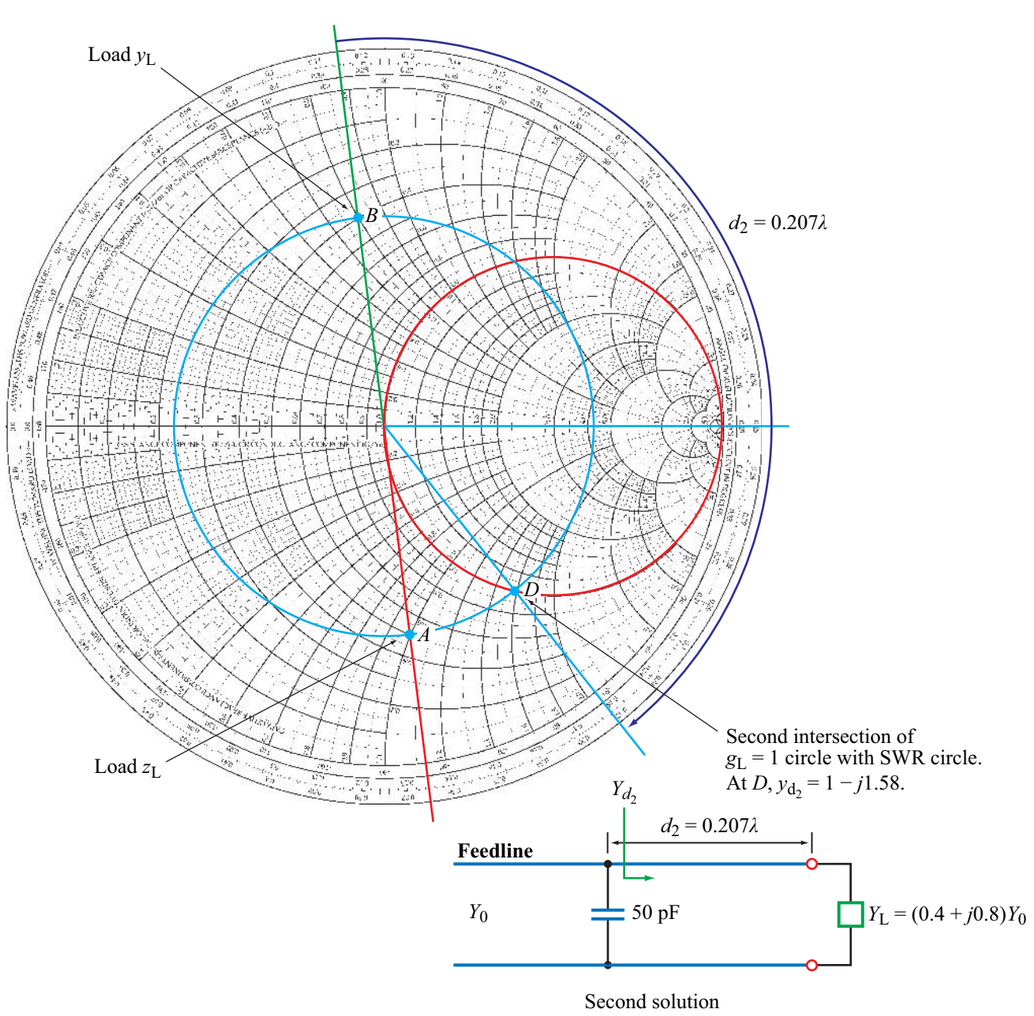


图 2-36 例 2-13 中 D 点的解法。点D是SWR圆和$g_L = 1$圆的第二个交点。 B 到 D 的距离为 $d_2 = 0.207\lambda$。